Six Most Important LITHIUM-ION BATTERY CHEMISTRIES

Li-ion cell is one of the most important energy storage devices in today's time. The cell has a lot of different chemistries, and the properties of these cells depend on their internal chemistry. Every single chemistry has its own pros and cons

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Lithium-ion (Li-ion) cells provide one of the most important energy storages today. These cells are used in many electronic items including smartphones, laptops, personal portable equipment, cameras, etc. In fact, Li-ion cells have paved the way for efficient and long ranged electromobility, thus making electrical vehicles (EVs) practical for our daily commute.

Li-ion cells are secondary cells or, in other words, rechargeable cells, that is, they can be recharged by passing current in the reverse direction. Li-ion batteries are lightweight and have higher energy density than other types of rechargeable cells, thus making them suitable for energy-intensive applications where constant replenishment of energy is essential, such as in our daily electronics gadgets, EVs, home energy storage system. These are used even in applications where a large amount of energy is stored, such as grid energy storage application.

This article describes the six most common Li-ion chemistries and the advantages as well as disadvantages of